Project Overview

The aim of this project was to simulate attacker behaviour against a vulnerable system and analyse the resulting logs using a Security Information and Event Management (SIEM) platform.

A Kali Linux virtual machine was used to perform reconnaissance and enumeration activities against a Metasploitable Linux server. Logs generated during these activities were collected and ingested into Splunk for investigation.

The objective was to demonstrate how a security analyst can identify suspicious activity by analysing authentication and web server logs.

Lab Environment Setup

The lab environment consisted of two virtual machines running on a Windows host.

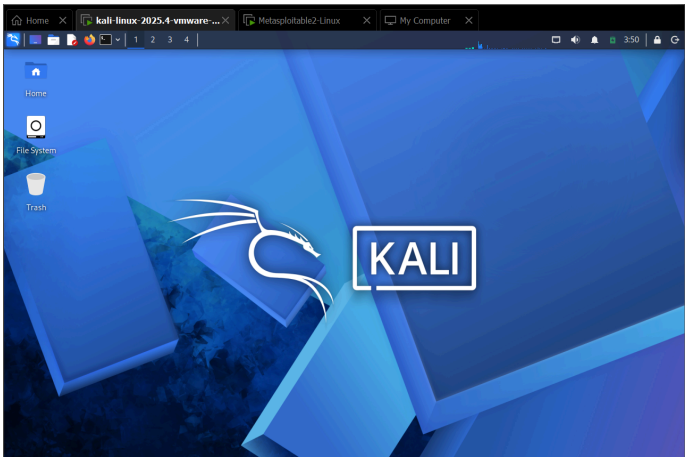
Kali Linux

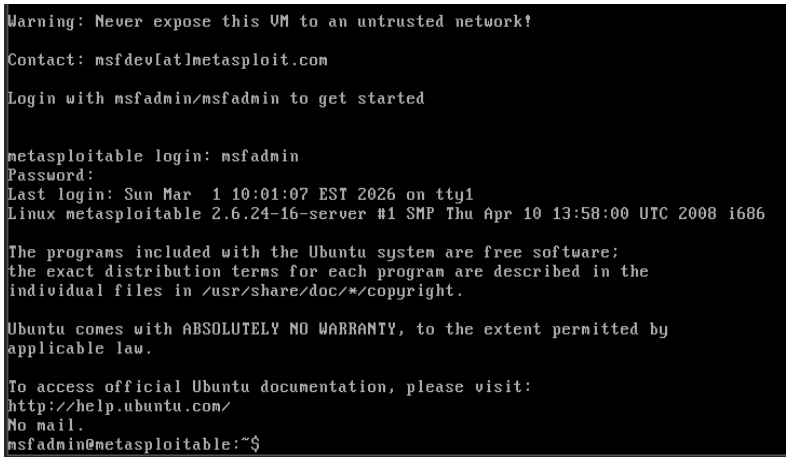
- Used as the attacker machine
- Performed reconnaissance and enumeration

Metasploitable 2

- A deliberately vulnerable Linux server
- Used as the target system

Both systems were configured on the same network to allow communication between attacker and target.





Network Configuration

Before performing any reconnaissance, the network connectivity between the systems was verified.

The IP address of the Kali Linux machine was identified.

```
┌──(kali㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
ault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 00:0c:29:24:e9:bf brd ff:ff:ff:ff:ff:ff
    inet 192.168.120.128/24 brd 192.168.120.255 scope global dynamic noprefix
route eth0
       valid_lft 1619sec preferred_lft 1619sec
    inet6 fe80::51bc:97eb:67d3:8ed8/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

The IP address of the Metasploitable system was then identified.

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:74:42:2d
          inet addr:192.168.120.129  Bcast:192.168.120.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe74:422d/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:284 errors:0 dropped:0 overruns:0 frame:0
          TX packets:100 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:22543 (22.0 KB)  TX bytes:13434 (13.1 KB)
          Interrupt:19 Base address:0x2000
```

A connectivity test was performed to ensure both machines could communicate.

```
┌──(kali㉿kali)-[~]
└─$ ping 192.168.120.129
PING 192.168.120.129 (192.168.120.129) 56(84) bytes of data.
64 bytes from 192.168.120.129: icmp_seq=1 ttl=64 time=0.520 ms
64 bytes from 192.168.120.129: icmp_seq=2 ttl=64 time=0.664 ms
64 bytes from 192.168.120.129: icmp_seq=3 ttl=64 time=2.42 ms
64 bytes from 192.168.120.129: icmp_seq=4 ttl=64 time=13.6 ms
^C
--- 192.168.120.129 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3050ms
rtt min/avg/max/mdev = 0.520/4.297/13.587/5.415 ms
```

The Windows host IP was also identified, as it would be used later for Splunk log analysis.

```
IPv4 Address. . . . . . . . . . . : 192.168.1.234
```

Connectivity between the host and virtual machines was verified.

```
┌──(kali㉿kali)-[~]
└─$ ping 192.168.1.234
PING 192.168.1.234 (192.168.1.234) 56(84) bytes of data.
64 bytes from 192.168.1.234: icmp_seq=1 ttl=128 time=3.17 ms
64 bytes from 192.168.1.234: icmp_seq=2 ttl=128 time=0.692 ms
64 bytes from 192.168.1.234: icmp_seq=3 ttl=128 time=0.957 ms
64 bytes from 192.168.1.234: icmp_seq=4 ttl=128 time=1.13 ms
^C
--- 192.168.1.234 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3033ms
rtt min/avg/max/mdev = 0.692/1.487/3.165/0.981 ms
```

```
msfadmin@metasploitable:~$ ping 192.168.1.234
PING 192.168.1.234 (192.168.1.234) 56(84) bytes of data.
64 bytes from 192.168.1.234: icmp_seq=1 ttl=128 time=10.7 ms
64 bytes from 192.168.1.234: icmp_seq=2 ttl=128 time=0.775 ms
64 bytes from 192.168.1.234: icmp_seq=3 ttl=128 time=0.788 ms
64 bytes from 192.168.1.234: icmp_seq=4 ttl=128 time=0.401 ms

--- 192.168.1.234 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 2999ms
rtt min/avg/max/mdev = 0.401/3.182/10.766/4.381 ms
msfadmin@metasploitable:~$ _
```

SIEM Setup

Splunk Enterprise was configured to ingest and analyse logs generated during the attack simulation.

A new index was created within Splunk to store the logs collected from the target system.

cyber-analyst-lab	Active	Events	search	1 MB	500 GB	0	–	–	$SPLUNK_DB\cyber-analyst-lab\db	–

The presence of system logs on the Metasploitable machine was confirmed.

```
msfadmin@metasploitable:~$ ls -l /var/log/auth.log
-rw-r--r-- 1 syslog adm 10235 2026-03-06 04:09 /var/log/auth.log
msfadmin@metasploitable:~$ ls -l /var/log/apache2/access.log
-rw-r--r-- 1 root root 183746 2026-03-01 10:38 /var/log/apache2/access.log
msfadmin@metasploitable:~$ _
```

Reconnaissance

Reconnaissance was performed using **Nmap** to identify open ports and services on the Metasploitable server.

Initial scanning identified several open services including:

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.120.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-06 04:19 -0500
Nmap scan report for 192.168.120.129
Host is up (0.0021s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 00:0C:29:74:42:2D (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.80 seconds
```

- SSH
- HTTP
- FTP
- Telnet

A more detailed scan was then performed to identify service versions.

nmap -sV <target-ip>

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV 192.168.120.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-06 04:23 -0500
Nmap scan report for 192.168.120.129
Host is up (0.0036s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login       OpenBSD or Solaris rlogind
514/tcp  open  tcpwrapped
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:74:42:2D (VMware)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:
linux:linux_kernel
```

Further enumeration was performed using the aggressive scan option.

nmap -A <target-ip>

```
┌──(kali㉿kali)-[~]
└─$ nmap -A 192.168.120.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-06 04:26 -0500
Nmap scan report for 192.168.120.129
Host is up (0.00078s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
```

These scans revealed multiple potentially vulnerable services running on the target system.

Attempted Exploitation

A brute force attack was attempted against the SSH service using Hydra.

hydra -l msfadmin -P rockyou.txt ssh://<target-ip>

```
┌──(kali㉿kali)-[~]
└─$ hydra -l msfadmin -P /usr/share/wordlists/rockyou.txt ssh://192.168.120.129
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service
 organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-06 04:34:11
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the ta
sks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:1/p:14344399), ~896525 tries
 per task
[DATA] attacking ssh://192.168.120.129:22/
```

The brute-force attempt was limited due to compatibility issues between modern SSH clients and the legacy configuration on Metasploitable.

Despite this limitation, authentication events were still generated within the system logs.

Web Enumeration

Directory enumeration was performed against the web server using Gobuster.

gobuster dir -u http://<target-ip> -w wordlist.txt

```
└─$ gobuster dir -u http://192.168.120.129 -w /usr/share/wordlists/dirb/common.txt
===============================================================
Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
===============================================================
[+] Url:                     http://192.168.120.129
[+] Method:                  GET
[+] Threads:                 10
[+] Wordlist:                /usr/share/wordlists/dirb/common.txt
[+] Negative Status codes:   404
[+] User Agent:              gobuster/3.8.2
[+] Timeout:                 10s
===============================================================
Starting gobuster in directory enumeration mode
===============================================================
.htpasswd            (Status: 403) [Size: 297]
.htaccess            (Status: 403) [Size: 297]
.hta                 (Status: 403) [Size: 292]
cgi-bin/             (Status: 403) [Size: 296]
dav                  (Status: 301) [Size: 321] [--> http://192.168.120.129/dav/]
index                (Status: 200) [Size: 891]
index.php            (Status: 200) [Size: 891]
phpMyAdmin           (Status: 301) [Size: 328] [--> http://192.168.120.129/phpMyAdmin/]
phpinfo              (Status: 200) [Size: 48029]
phpinfo.php          (Status: 200) [Size: 48041]
server-status        (Status: 403) [Size: 301]
test                 (Status: 301) [Size: 322] [--> http://192.168.120.129/test/]
twiki                (Status: 301) [Size: 323] [--> http://192.168.120.129/twiki/]
Progress: 4613 / 4613 (100.00%)
===============================================================
Finished
===============================================================
```

The scan revealed several directories and files including:

/phpinfo.php
/phpMyAdmin
/test
/twiki

These directories could potentially expose sensitive information or provide entry points for further attacks.

```
msfadmin@metasploitable:~$ ls -l /var/log/apache2/access.log
-rw-r--r-- 1 root root 183746 2026-03-01 10:38 /var/log/apache2/access.log
msfadmin@metasploitable:~$ tail /var/log/auth.log
Mar  6 04:25:57 metasploitable sshd[5382]: Did not receive identification string
 from 192.168.120.128
Mar  6 04:26:03 metasploitable rlogind[5392]: Connection from 192.168.120.128 on
 illegal port
Mar  6 04:26:03 metasploitable rlogind[5407]: Connection from 192.168.120.128 on
 illegal port
Mar  6 04:26:03 metasploitable rlogind[5418]: Connection from 192.168.120.128 on
 illegal port
Mar  6 04:26:09 metasploitable sshd[5423]: Protocol major versions differ for 19
2.168.120.128: SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1 vs. SSH-1.5-Nmap-SSH1-Hostk
ey
Mar  6 04:26:09 metasploitable sshd[5422]: Protocol major versions differ for 19
2.168.120.128: SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1 vs. SSH-1.5-NmapNSE_1.0
Mar  6 04:26:09 metasploitable rshd[5449]: Connection from 192.168.120.128 on il
legal port
Mar  6 04:26:14 metasploitable rshd[5494]: Connection from 192.168.120.128 on il
legal port
Mar  6 04:39:01 metasploitable CRON[5555]: pam_unix(cron:session): session opene
d for user root by (uid=0)
Mar  6 04:39:01 metasploitable CRON[5555]: pam_unix(cron:session): session close
d for user root
msfadmin@metasploitable:~$ _
```

```
msfadmin@metasploitable:~$ tail /var/log/apache2/access.log
192.168.120.128 - - [06/Mar/2026:04:38:12 -0500] "GET /zt HTTP/1.1" 404 286 "-"
"gobuster/3.8.2"
192.168.120.128 - - [06/Mar/2026:04:38:12 -0500] "GET /zoom HTTP/1.1" 404 288 "-
" "gobuster/3.8.2"
192.168.120.128 - - [06/Mar/2026:04:38:12 -0500] "GET /zope HTTP/1.1" 404 288 "-
" "gobuster/3.8.2"
192.168.120.128 - - [06/Mar/2026:04:38:12 -0500] "GET /zorum HTTP/1.1" 404 289 "
-" "gobuster/3.8.2"
192.168.120.128 - - [06/Mar/2026:04:38:12 -0500] "GET /worldpayreturn HTTP/1.1"
404 298 "-" "gobuster/3.8.2"
192.168.120.128 - - [06/Mar/2026:04:38:12 -0500] "GET /ws_ftp HTTP/1.1" 404 290
"-" "gobuster/3.8.2"
192.168.120.128 - - [06/Mar/2026:04:38:13 -0500] "GET /zap HTTP/1.1" 404 287 "-"
 "gobuster/3.8.2"
127.0.0.1 - - [06/Mar/2026:04:38:14 -0500] "OPTIONS * HTTP/1.0" 200 - "-" "Apach
e/2.2.8 (Ubuntu) DAV/2 (internal dummy connection)"
127.0.0.1 - - [06/Mar/2026:04:38:15 -0500] "OPTIONS * HTTP/1.0" 200 - "-" "Apach
e/2.2.8 (Ubuntu) DAV/2 (internal dummy connection)"
127.0.0.1 - - [06/Mar/2026:04:38:16 -0500] "OPTIONS * HTTP/1.0" 200 - "-" "Apach
e/2.2.8 (Ubuntu) DAV/2 (internal dummy connection)"
msfadmin@metasploitable:~$
```

Log Collection

System logs generated during the attack were collected from the Metasploitable system.

The key logs identified for analysis were:

- auth.log
- access.log

```
msfadmin@metasploitable:~$ cd /var/log
msfadmin@metasploitable:/var/log$ cp auth.log /home/msfadmin/
msfadmin@metasploitable:/var/log$ cp apache2/access.log /home/msfadmin
msfadmin@metasploitable:/var/log$ ls /home/msfadmin
access.log  auth.log  vulnerable
msfadmin@metasploitable:/var/log$ _
```

An attempt was made to transfer the logs using SCP.

```
┌──(kali㉿kali)-[~]
└─$ scp msfadmin@192.168.120.129:/home/msfadmin/access.log .
Unable to negotiate with 192.168.120.129 port 22: no matching host key type found. Their offer: ssh-rsa,
ssh-dss
scp: Connection closed
```

Due to compatibility issues with the legacy SSH configuration, this method was unsuccessful.

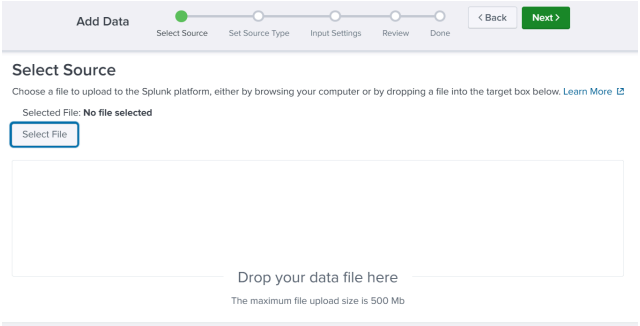
As an alternative, the logs were retrieved via HTTP.

```
┌──(kali㉿kali)-[~]
└─$ wget http://192.168.120.129/auth.log
--2026-03-06 05:13:21--  http://192.168.120.129/auth.log
Connecting to 192.168.120.129:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12278 (12K) [text/plain]
Saving to: 'auth.log'

auth.log                100%[===================================>]  11.99K  --.-KB/s    in 0s

2026-03-06 05:13:21 (643 MB/s) - 'auth.log' saved [12278/12278]


┌──(kali㉿kali)-[~]
└─$ wget http://192.168.120.129/access.log
--2026-03-06 05:13:34--  http://192.168.120.129/access.log
Connecting to 192.168.120.129:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 659967 (644K) [text/plain]
Saving to: 'access.log'

access.log              100%[===================================>] 644.50K  --.-KB/s    in 0.01s

2026-03-06 05:13:34 (46.2 MB/s) - 'access.log' saved [659967/659967]
```

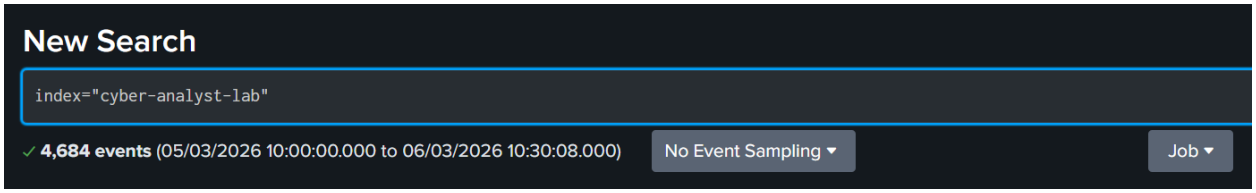
The logs were then prepared for transfer to the host system by using drag and drop from the Linux Machine to the Windows Desktop.

Log Ingestion into Splunk

The collected logs were uploaded into Splunk using the **Add Data** function.



After ingestion, Splunk confirmed that thousands of events were successfully indexed.



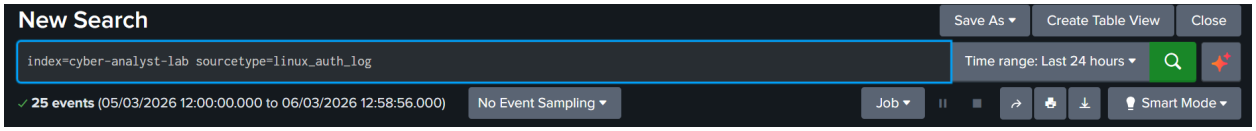
This confirmed that the SIEM platform could process and analyse the log data.

Authentication Log Analysis

Authentication logs were analysed to identify suspicious login or connection attempts.

The following query was used:

index=cyber-analyst-lab sourcetype=linux_auth_log



Several authentication-related events were identified during the investigation.

A couple of events showed suspicious connection attempts from the attacking machine.

```
06/03/2026 04:26:14.000    JamesLaptop    auth.log.log    linux_auth_log

Mar  6 04:26:14 metasploitable rshd[5494]: Connection from 192.168.120.128 on illegal port
```

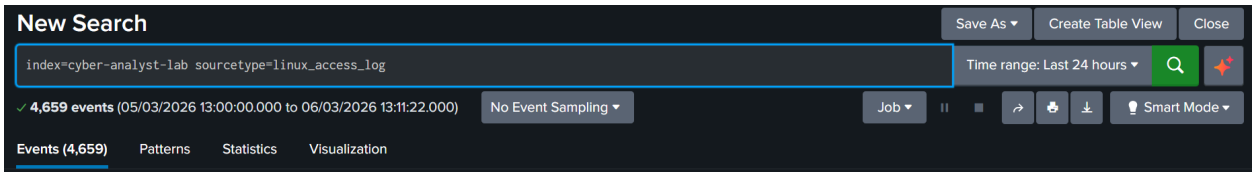
This event indicates that the attacking host attempted to interact with a legacy remote shell service.

Web Server Log Analysis

Apache access logs were analysed to identify suspicious web requests generated during enumeration.

The following query was used:

index=cyber-analyst-lab sourcetype=linux_access_log



The logs revealed several directory enumeration attempts originating from the Kali machine.

Examples included:

GET /phpinfo.php
GET /phpMyAdmin
GET /test
GET /twiki

The user agent recorded in the logs identified the scanning tool.

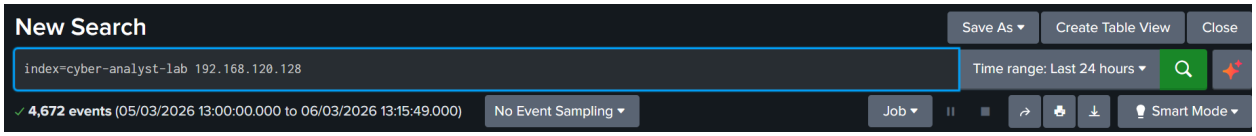
gobuster/3.8.2

This confirmed that automated directory enumeration was performed against the web server.

Identifying the Attacker

To identify the source of the suspicious activity, searches were performed for the attacking machine's IP address.

index=cyber-analyst-lab 192.168.120.128



The logs clearly show multiple requests originating from the Kali system, confirming that the activity was generated during the simulated attack.

Findings

The investigation revealed several key indicators of suspicious activity:

1. Authentication Events

- Connection attempts recorded in authentication logs
- Activity originating from the attacking host

2. Web Enumeration Activity

- Multiple HTTP requests to hidden directories
- Automated scanning behaviour detected

3. Attacker Identification

- Logs confirmed that the source IP address for the activity was the Kali attacker machine

Attack Timeline

Reconnaissance- Nmap scan identified open services

Enumeration- Gobbuster discovered hidden web directories

Exploitation Attempt- Hydra attempted SSH brute force

Log Collection- auth.log and access.log collected

SIEM Analysis- Logs ingested and investigated in Splunk

Indicators of Compromise

Suspicious user-agent

- gobuster/3.8.2

Suspicious requests

```
06/03/2026 09:38:11.000   192.168.120.128 - - [06/Mar/2026:04:38:11 -0500] "GET /phpMyAdmin HTTP/1.1" 301 328 "-" "gobuster/3.8.2"
host = JamesLaptop | source = access.log.log | sourcetype = linux_access_log

06/03/2026 09:38:12.000   192.168.120.128 - - [06/Mar/2026:04:38:12 -0500] "GET /test HTTP/1.1" 301 322 "-" "gobuster/3.8.2"
host = JamesLaptop | source = access.log.log | sourcetype = linux_access_log

06/03/2026 09:38:12.000   192.168.120.128 - - [06/Mar/2026:04:38:12 -0500] "GET /twiki HTTP/1.1" 301 323 "-" "gobuster/3.8.2"
host = JamesLaptop | source = access.log.log | sourcetype = linux_access_log

06/03/2026 09:38:11.000   192.168.120.128 - - [06/Mar/2026:04:38:11 -0500] "GET /phpinfo.php HTTP/1.1" 200 48041 "-" "gobuster/3.8.2"
host = JamesLaptop | source = access.log.log | sourcetype = linux_access_log

06/03/2026 09:38:11.000   192.168.120.128 - - [06/Mar/2026:04:38:11 -0500] "GET /phpinfo HTTP/1.1" 200 48029 "-" "gobuster/3.8.2"
host = JamesLaptop | source = access.log.log | sourcetype = linux_access_log
```

Conclusion

This project demonstrated how attacker activity can be detected through log analysis using a SIEM platform.

By simulating reconnaissance and enumeration attacks within a controlled lab environment, logs were generated that provided evidence of suspicious behaviour.

Using Splunk, these logs were successfully ingested and analysed to identify authentication events, web enumeration activity, and the source of the attack.

This investigation highlights the importance of log monitoring and SIEM tools in identifying and responding to potential security incidents.